

Implement Iris Flower Classification using KNN

CODE:

```
# Iris Flower Classification using KNN
```

```
from sklearn.datasets import load_iris
```

```
from sklearn.model_selection import train_test_split
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```
from sklearn.neighbors import KNeighborsClassifier
```

```
from sklearn.metrics import accuracy_score, confusion_matrix
```

```
# Load Iris dataset
```

```
iris = load_iris()
```

```
X = iris.data
```

```
y = iris.target
```

```
# Split dataset into training and testing data
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```
X_train, X_test, y_train, y_test = train_test_split(
```

```
    X, y, test_size=0.2, random_state=42
```

```
)
```

```
# Create KNN model
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```
k = 5
```

```
model = KNeighborsClassifier(n_neighbors=k)
```

```
# Train the model
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```
model.fit(X_train, y_train)
```

```
# Predict test data
y_pred = model.predict(X_test)

# Calculate accuracy
accuracy = accuracy_score(y_test, y_pred)

# Display results
print("Iris Flower Classification using KNN")
print("-----")

print("K Value:", k)

print("\nAccuracy:", accuracy)

print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred))

# Predict a new flower
new_flower = [[5.1, 3.5, 1.4, 0.2]]
prediction = model.predict(new_flower)

print("\nNew Flower Measurements:")
print(new_flower)

print("Predicted Flower:",
      iris.target_names[prediction[0]])
```

OUTPUT:

```
*** Iris Flower Classification using KNN
-----
K Value: 5

Accuracy: 1.0

Confusion Matrix:
[[10  0  0]
 [ 0  9  0]
 [ 0  0 11]]

New Flower Measurements:
[[5.1, 3.5, 1.4, 0.2]]
Predicted Flower: setosa
```